# Network Concept of Security

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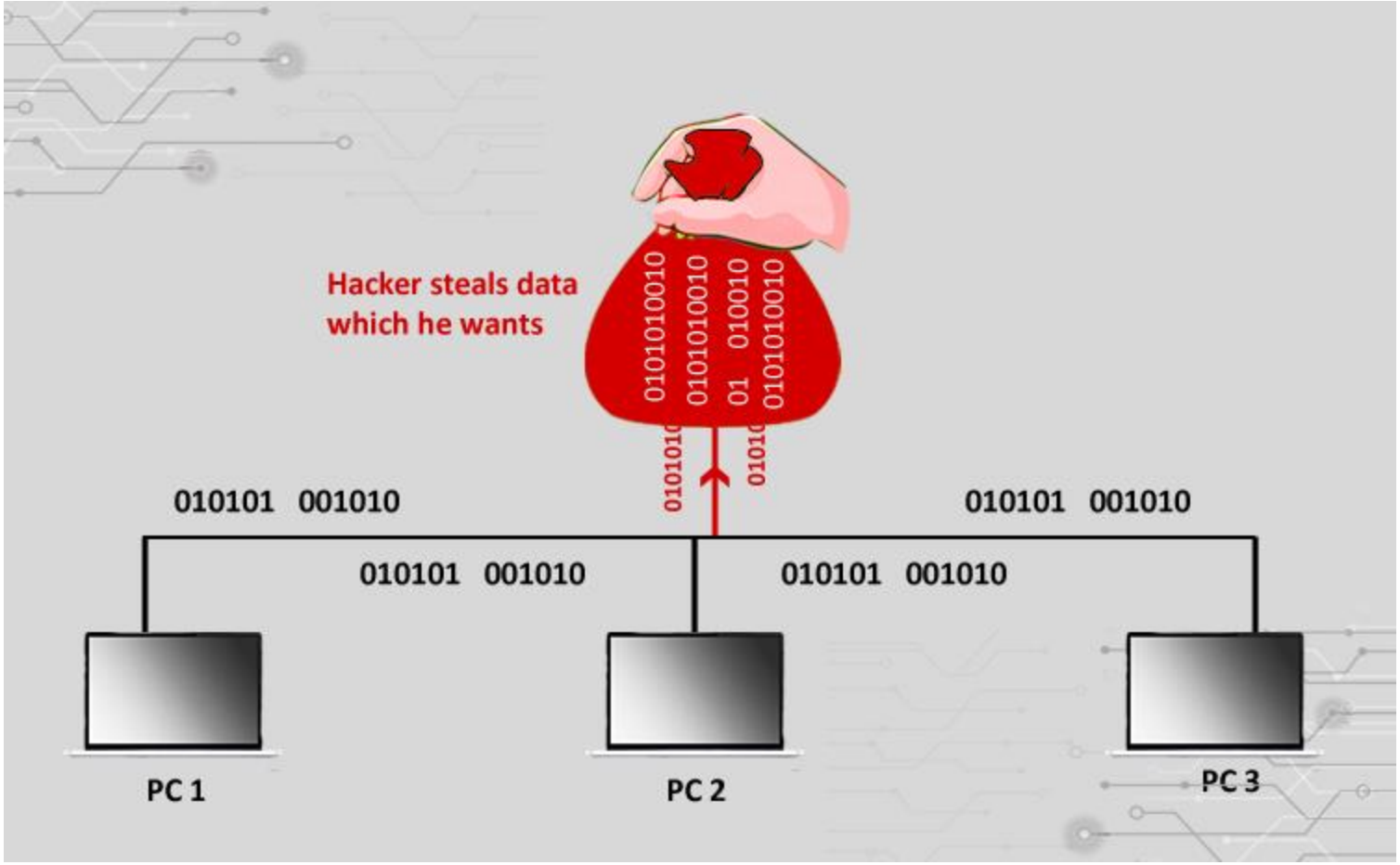
# Unit - 4

## Wireless Attacks

# Network Sniffing

Network sniffing is the process of capturing and analyzing network traffic for various purposes, such as troubleshooting, monitoring, or, unfortunately, unauthorized interception of sensitive information. It involves capturing packets of data as they travel across a computer network.


Hacker steals data
which he wants
0101010010
0101010010
01 010010
0101010010
010101 001010
010101 001010
010101 001010
010101 001010
PC 1
PC 2
PC 3

# Purpose of Network Sniffing:

Troubleshooting: Network administrators use sniffing tools to diagnose and troubleshoot network issues. Analyzing packet data can help identify the source of problems such as network congestion, errors, or security issues.

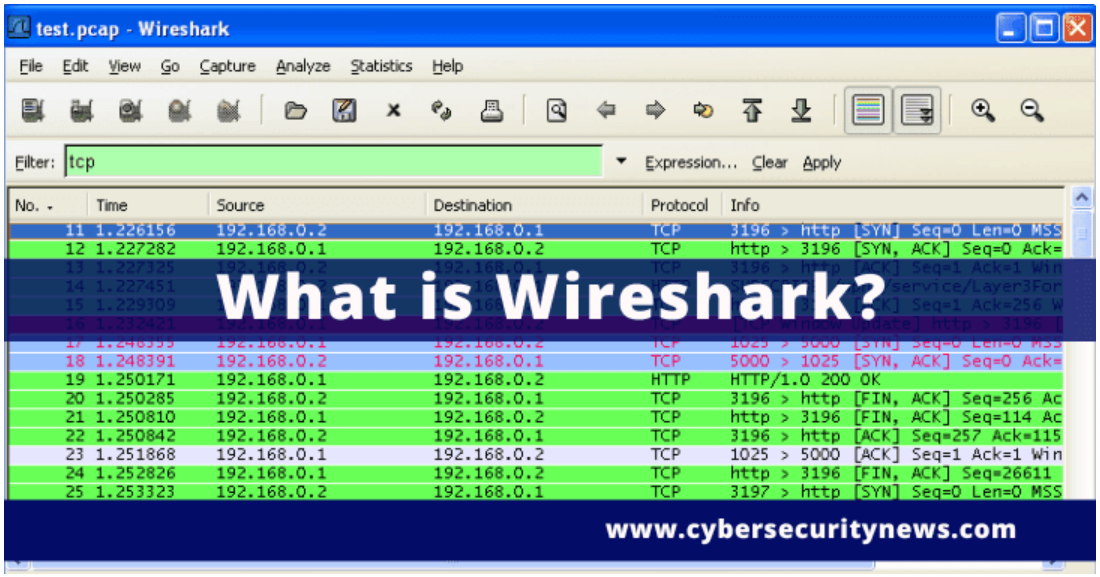
Security Monitoring: Security professionals use network sniffing to detect and analyze suspicious or malicious activities on a network. This can include identifying unauthorized access, malicious traffic patterns, or potential security threats.

Performance Monitoring: Sniffing tools can be used to monitor and analyze network performance, helping organizations optimize their network infrastructure for efficiency.

Protocol Analysis: Network sniffers allow for the examination of the protocols used in network communication. This is useful for understanding how different applications and devices communicate on the network.

# Wireshark

Wireshark is one of the most popular and widely used network protocol analyzers. It allows users to capture and analyze the data traveling back and forth on a network in real-time.


test.pcap - Wireshark
File Edit View Go Capture Analyze Statistics Help
Filter: tcp
Expression... Clear Apply
No. Time Source Destination Protocol Info
11 1.226156 192.168.0.2 192.168.0.1 TCP 3196 > http [SYN] Seq=0 Len=0 MSS
12 1.227282 192.168.0.1 192.168.0.2 TCP http > 3196 [SYN, ACK] Seq=0 Ack=
14 1.227451
15 1.229309
18 1.248391 192.168.0.2 192.168.0.1 TCP 5000 > 1025 [SYN, ACK] Seq=0 Ack=
19 1.250171 192.168.0.1 192.168.0.2 HTTP HTTP/1.0 200 OK
20 1.250285 192.168.0.2 192.168.0.1 TCP 3196 > http [FIN, ACK] Seq=256 Ac
21 1.250810 192.168.0.1 192.168.0.2 TCP http > 3196 [FIN, ACK] Seq=114 Ac
22 1.250842 192.168.0.2 192.168.0.1 TCP 3196 > http [ACK] Seq=257 Ack=115
23 1.251868 192.168.0.1 192.168.0.2 TCP 1025 > 5000 [ACK] Seq=1 Ack=1 Win
24 1.252826 192.168.0.1 192.168.0.2 TCP http > 3196 [FIN, ACK] Seq=26611
25 1.253323 192.168.0.2 192.168.0.1 TCP 3197 > http [SYN] Seq=0 Len=0 MSS
What is Wireshark?
www.cybersecuritynews.com

# Features:

Packet Capture: Wireshark captures packets on a network in real-time or reads previously saved capture files.

Protocol Analysis: It supports a wide range of network protocols, allowing users to analyze the details of each packet and understand the communication between devices.

Live Capture and Offline Analysis: Wireshark can capture live data as it is transmitted over a network or analyze pre-recorded capture files.

Filtering and Search: Users can apply filters to focus on specific types of packets or communication. This makes it easier to locate and analyze relevant data.

Colorization and Decoding: Wireshark colorizes packets based on the type of protocol, making it visually intuitive. It also decodes packets to display human-readable information.

Statistics and Graphs: The application provides statistical information about captured data, and it can generate various graphs to illustrate network trends.

VoIP Analysis: Wireshark has features specifically designed for Voice over Internet Protocol (VoIP) analysis, allowing users to examine SIP, RTP, and other VoIP protocols.

Exporting Data: Captured data can be exported in various formats for further analysis or reporting.

# Packet analysis

Packet analysis in networks involves the examination and interpretation of individual data packets that traverse a computer network. This process is essential for understanding network behavior, troubleshooting issues, monitoring performance, and identifying security threats.

tv-netflix-problems-2011-07-06.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression... +

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 343 | 65.142415 | 192.168.0.21 | 174.129.249.228 | TCP | 66 | 40555 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519346 TSecr=551811827 |
| 344 | 65.142715 | 192.168.0.21 | 174.129.249.228 | HTTP | 253 | GET /clients/netflix/flash/application.swf?flash_version=flash_lite_2.1&v=1.5&nr |
| 345 | 65.230738 | 174.129.249.228 | 192.168.0.21 | TCP | 66 | 80 → 40555 [ACK] Seq=1 Ack=188 Win=6864 Len=0 TSval=551811850 TSecr=491519347 |
| 346 | 65.240742 | 174.129.249.228 | 192.168.0.21 | HTTP | 828 | HTTP/1.1 302 Moved Temporarily |
| 347 | 65.241592 | 192.168.0.21 | 174.129.249.228 | TCP | 66 | 40555 → 80 [ACK] Seq=188 Ack=763 Win=7424 Len=0 TSval=491519446 TSecr=551811852 |
| 348 | 65.242532 | 192.168.0.21 | 192.168.0.1 | DNS | 77 | Standard query 0x2188 A cdn-0.nflximg.com |
| 349 | 65.276870 | 192.168.0.1 | 192.168.0.21 | DNS | 489 | Standard query response 0x2188 A cdn-0.nflximg.com CNAME images.netflix.com.edge |
| 350 | 65.277992 | 192.168.0.21 | 63.80.242.48 | TCP | 74 | 37063 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=491519482 TSecr |
| 351 | 65.297757 | 63.80.242.48 | 192.168.0.21 | TCP | 74 | 80 → 37063 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM=1 TSval=3295 |
| 352 | 65.298396 | 192.168.0.21 | 63.80.242.48 | TCP | 66 | 37063 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519502 TSecr=3295534130 |
| 353 | 65.298687 | 192.168.0.21 | 63.80.242.48 | HTTP | 153 | GET /us/nrd/clients/flash/814540.bun HTTP/1.1 |
| 354 | 65.318730 | 63.80.242.48 | 192.168.0.21 | TCP | 66 | 80 → 37063 [ACK] Seq=1 Ack=88 Win=5792 Len=0 TSval=3295534151 TSecr=491519503 |
| 355 | 65.321733 | 63.80.242.48 | 192.168.0.21 | TCP | 1514 | [TCP segment of a reassembled PDU] |

```
> Frame 349: 489 bytes on wire (3912 bits), 489 bytes captured (3912 bits)
> Ethernet II, Src: Globalsc_00:3b:0a (f0:ad:4e:00:3b:0a), Dst: Vizio_14:8a:e1 (00:19:9d:14:8a:e1)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.21
> User Datagram Protocol, Src Port: 53 (53), Dst Port: 34036 (34036)
v Domain Name System (response)
    [Request In: 348]
    [Time: 0.034338000 seconds]
    Transaction ID: 0x2188
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 4
    Authority RRs: 9
    Additional RRs: 9
  v Queries
    > cdn-0.nflximg.com: type A, class IN
  > Answers
  > Authoritative nameservers
```

```
0020  00 15 00 35 84 f4 01 c7  83 3f 21 88 81 80 00 01   ...5.... .?!.....
0030  00 04 00 09 00 09 05 63  64 6e 2d 30 07 6e 66 6c   .......c dn-0.nfl
0040  78 69 6d 67 03 63 6f 6d  00 00 01 00 01 c0 0c 00   ximg.com ........
0050  05 00 01 00 00 05 29 00  22 06 69 6d 61 67 65 73   ......). ".images
0060  07 6e 65 74 66 6c 69 78  03 63 6f 6d 09 65 64 67   .netflix .com.edg
0070  65 73 75 69 74 65 03 6e  65 74 00 c0 2f 00 05 00   esuite.n et../...
```

Identification of transaction (dns.id), 2 bytes | Packets: 10299 · Displayed: 10299 (100.0%) · Load time: 0:0.182 | Profile: Default

# Aspects of packet analysis:

Capture Packets:
Use packet capture tools like Wireshark to capture network traffic in real-time or analyze saved capture files.
Choose the appropriate network interface for capturing data.

Packet Structure:
Understand the structure of network packets, which typically includes headers and payloads.
Different protocols have different packet structures.

Protocol Analysis:
Identify the protocols used in the communication (e.g., TCP, UDP, IP, HTTP, DNS).
Analyze protocol-specific details such as flags, sequence numbers, and control information.

Packet Details:
Examine packet details to understand source and destination addresses, port numbers, and timing information.
Investigate the contents of packet payloads for application-specific data.

Filtering:
Apply filters to focus on specific types of packets, IP addresses, or communication flows.
Filtering helps in narrowing down the analysis to relevant data.

Follow Streams:
Use features like "Follow TCP Stream" in Wireshark to reconstruct and view entire conversations between network nodes.
This helps in understanding the context of communication.

Security Analysis:
Look for signs of security threats, such as unusual traffic patterns, suspicious IP addresses, or unauthorized access attempts.
Identify and analyze anomalies that may indicate potential security incidents.

VoIP Analysis:
If applicable, analyze Voice over Internet Protocol (VoIP) traffic for quality issues, call setup problems, or other issues. VoIP protocols like SIP and RTP have specific characteristics to consider.

Statistical Analysis:
Utilize statistical tools within packet analysis software to gather insights into network utilization, error rates, and performance metrics.
Graphs and charts can help visualize trends.

Encryption Analysis:
Recognize encrypted traffic and understand that the payload may be unreadable without the encryption keys.
Note the use of secure protocols like HTTPS.

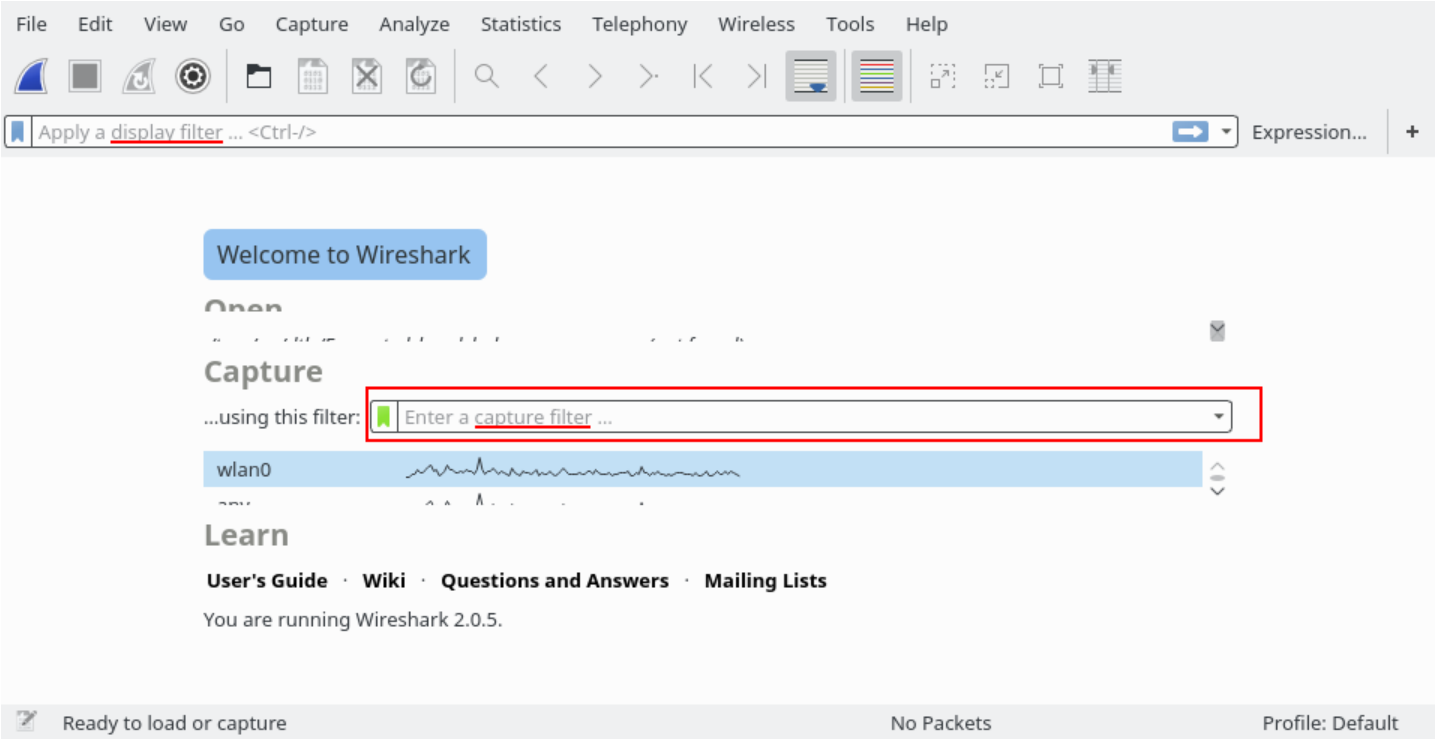
Troubleshooting:
Packet analysis is a powerful tool for troubleshooting network issues. Identify and address problems such as retransmissions, dropped packets, or misconfigurations.

Documentation:
Document findings and observations for future reference or to share information with colleagues.
Annotations in packet capture tools help in providing context to the analysis.

Educational Resources:
Continuously educate yourself on network protocols, packet structures, and evolving technologies to enhance your packet analysis skills.

# Display and capture filters

In network packet analysis, filters are used to selectively display or capture specific packets based on certain criteria. Filters help narrow down the focus of analysis to relevant information and are commonly employed in tools like Wireshark.


File
Edit
View
Go
Capture
Analyze
Statistics
Telephony
Wireless
Tools
Help
Apply a display filter ... <Ctrl-/>
Expression...
+
Welcome to Wireshark
Capture
...using this filter:
Enter a capture filter ...
wlan0
Learn
User's Guide · Wiki · Questions and Answers · Mailing Lists
You are running Wireshark 2.0.5.
Ready to load or capture
No Packets
Profile: Default

# Display Filters

Display filters are applied after packets have been captured, allowing you to selectively view and analyze specific packets within a packet capture file.

# Examples

Protocol Filters:
Display only TCP packets: tcp
Display only UDP packets: udp
Display only ICMP packets: icmp

Address Filters:
Display packets to or from a specific IP address: ip.addr == 192.168.1.1
Display packets within a specific IP range: ip.addr in 192.168.1.0/24

# Capture Filters

Capture filters are applied during the packet capture process, allowing you to capture only the packets that meet specific criteria. Capture filters are specified before starting the packet capture.

# Examples

Host Filters:
Capture packets to or from a specific host: host 192.168.1.1
Capture packets within a specific IP range: net 192.168.1.0/24

Port Filters:
Capture packets with a specific source or destination port: port 80 (HTTP)
Capture packets with source or destination port range: portrange 5000-5010

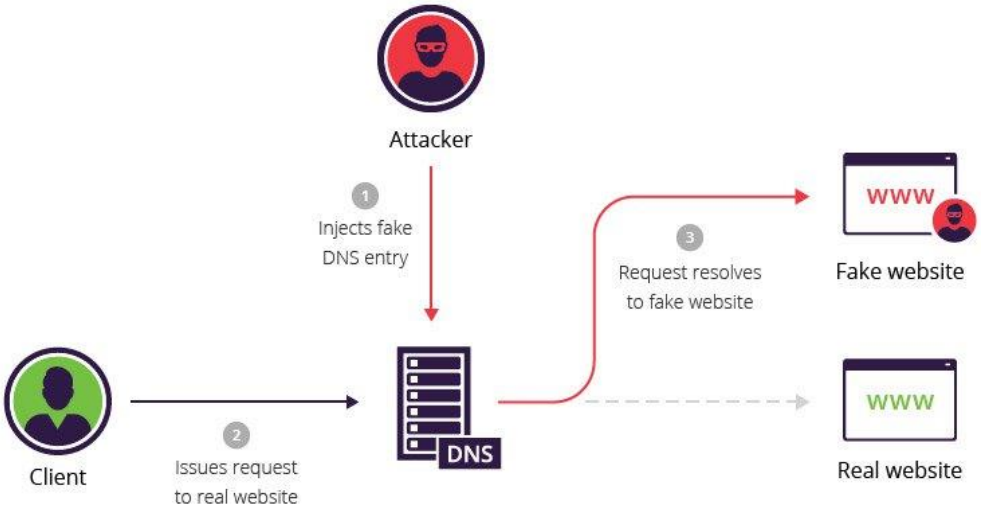
# Ettercap

Ettercap is a comprehensive suite for man-in-the-middle attacks on computer networks. It is an open-source tool designed to perform various network security tasks, including network sniffing, protocol analysis, and active and passive attacks. Ettercap supports both Ethernet and WiFi networks and is available for Unix/Linux and Windows platforms.

Ettercap
0.8.3
Setup
Sniffing at startup
Primary Interface
eth0
Bridged sniffing
Bridged Interface
Local Loopback
Ettercap

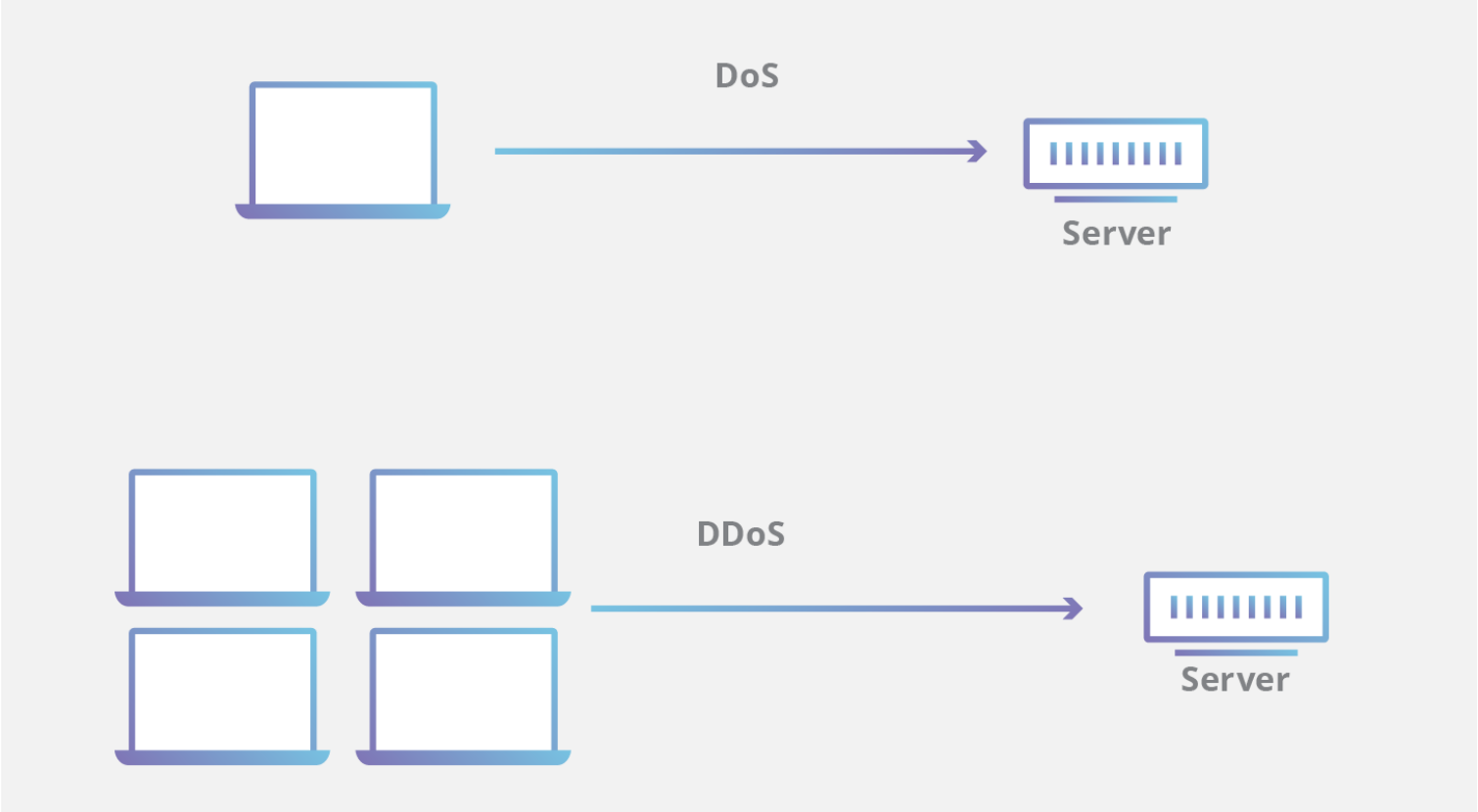
# DNS poisoning

DNS (Domain Name System) poisoning, also known as DNS spoofing, is a type of cyber attack that involves manipulating the DNS resolution process to redirect domain name queries to malicious IP addresses. The goal of DNS poisoning is typically to mislead users and divert their traffic to a fraudulent website, allowing attackers to intercept sensitive information or conduct other malicious activities.


Attacker
1
Injects fake
DNS entry
3
Request resolves
to fake website
www
Fake website
Client
2
Issues request
to real website
DNS
www
Real website

# Denial of Service (DoS) attack

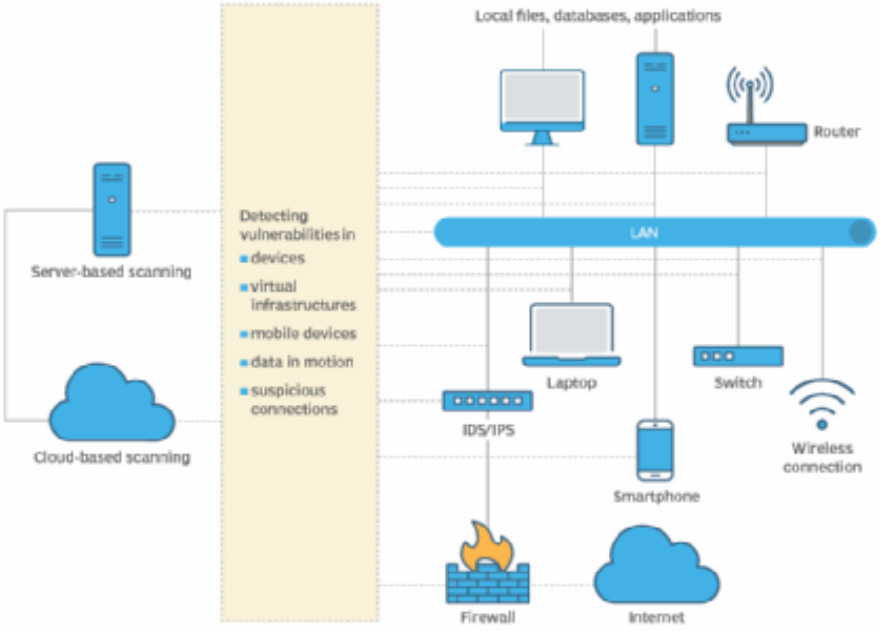
A Denial of Service (DoS) attack is a malicious attempt to disrupt the normal functioning of a targeted server, service, or network by overwhelming it with a flood of traffic. The primary objective of a DoS attack is to make the targeted resource unavailable to its intended users.


DoS
Server
DDoS
Server

# Vulnerability scanning

Vulnerability scanning is a proactive approach to identify and assess potential security vulnerabilities in a network, system, or application. It involves using automated tools to systematically discover and analyze weaknesses that could be exploited by attackers.

# How a vulnerability scanner works



# Key Components of Vulnerability Scanning:

Network Discovery:
The scanning tool identifies and maps the network, including active hosts, devices, and services.

Vulnerability Detection:
Automated tools scan for known vulnerabilities in operating systems, applications, and services running on networked devices.

Risk Assessment:
Each identified vulnerability is assigned a risk level based on factors such as the severity of the vulnerability and the potential impact on the organization.

Reporting:
A detailed report is generated, providing information about discovered vulnerabilities, their severity, and recommendations for remediation.

# Nessus

Nessus is a widely used and powerful vulnerability scanning tool designed to identify security vulnerabilities, misconfigurations, and other weaknesses in networks, systems, and applications. Developed by Tenable, Nessus is known for its extensive vulnerability database and its ability to perform thorough and accurate security assessments.

# Vulnerability Scanner

# Create Policy

# Features:

Vulnerability Scanning:
Nessus performs automated vulnerability scans on networks, systems, and applications to identify potential security issues.

Plugin Architecture:
The strength of Nessus lies in its extensive plugin architecture. It supports a large number of plugins that cover a wide range of vulnerabilities, including those related to operating systems, services, and applications.

# Features:

Compliance Checks:
Nessus includes compliance checks against various security standards and regulatory frameworks (such as PCI DSS, HIPAA) to help organizations ensure compliance with industry-specific requirements.

Scanning Profiles:
Users can customize scanning profiles based on their specific needs, adjusting the level of aggressiveness, scanning scope, and other parameters.

Policy Auditing:
Nessus can audit systems against security policies, providing insights into security configurations and compliance.

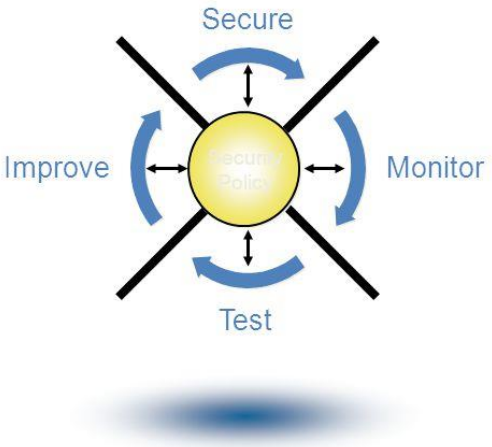
Web Application Scanning:
In addition to network scanning, Nessus includes web application scanning capabilities to identify vulnerabilities in web applications and services.

# Network policies

Network policies are sets of rules and guidelines that define how network resources and services should be accessed, used, and secured within an organization. These policies play a crucial role in establishing the framework for network management, security, and compliance.

# Network Security as a Continuous Process

- Network security is a continuous process built around a security policy.
  - Step 1: Secure
  - Step 2: Monitor
  - Step 3: Test
  - Step 4: Improve



# Network policies:

Access Control Policies:
Define rules for controlling access to network resources based on user roles, job responsibilities, and the principle of least privilege.
Specify authentication methods, authorization levels, and conditions for granting or denying access.

# Network policies:

Firewall Policies:
Establish rules for the configuration and management of firewalls to control the flow of traffic between different network segments.
Define policies for allowing or blocking specific types of traffic, based on protocols, ports, and IP addresses.

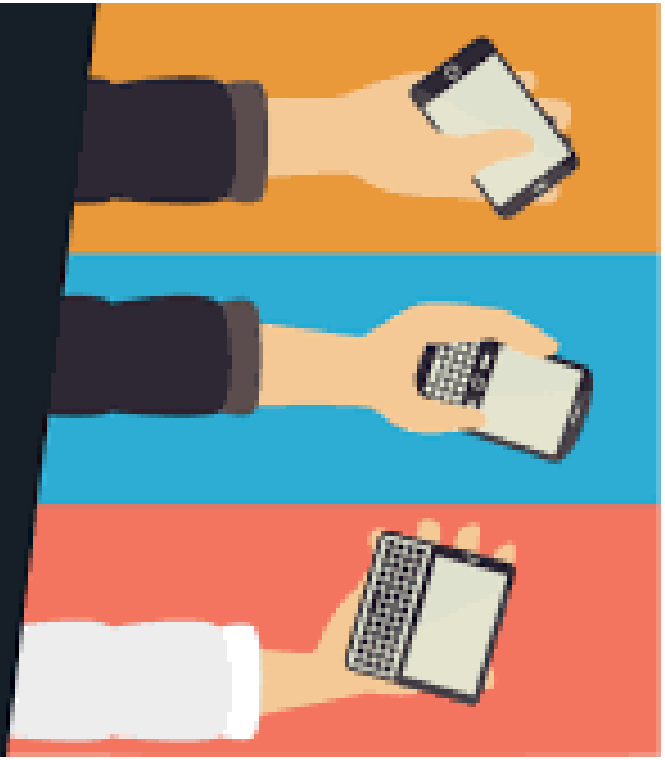
Security Policies:
Outline security measures and practices to protect the network from unauthorized access, attacks, and data breaches.
Include guidelines for encryption, intrusion detection and prevention, antivirus measures, and incident response.

BYOD (Bring Your Own Device) Policies:
Specify rules for the use of personal devices (such as smartphones and laptops) on the corporate network. Address security concerns, device registration, and acceptable use practices for personal devices.

# Bring Your Own Device (BYOD)

*The policy by which employees can use their own personal mobile devices to access company networks*

ACCOUNTANTS etc



Remote Access Policies:

Define procedures and security measures for remote access to the network, especially for employees working outside the office.

Specify the use of Virtual Private Networks (VPNs), multi-factor authentication, and encryption.

Network Monitoring and Logging Policies:
Establish guidelines for monitoring network activities and maintaining logs for security and compliance purposes. Define the types of events to be logged, retention periods, and procedures for reviewing logs.

Data Classification and Handling Policies:
Categorize data based on sensitivity and importance.
Define rules for handling, transmitting, and storing data, considering its classification.

Incident Response and Reporting Policies:
Outline procedures for responding to security incidents, including reporting, containment, eradication, recovery, and lessons learned.
Establish communication protocols during and after incidents.

# Importance of Network Policies:

Security and Risk Mitigation: Network policies are essential for identifying and mitigating security risks, ensuring the confidentiality, integrity, and availability of data.

Compliance: Policies help organizations comply with industry regulations, legal requirements, and standards relevant to their operations.

Consistency: Establishing and enforcing network policies promotes consistency in network management, reducing the likelihood of errors and vulnerabilities.

Efficient Network Operations: Well-defined policies contribute to the efficient and organized operation of network resources, promoting a secure and reliable computing environment.

Employee Awareness: Policies serve as a guide for employees, helping them understand their responsibilities, acceptable practices, and security measures.

Incident Response: Clearly defined policies expedite the response to security incidents, minimizing the impact and facilitating a faster recovery.

Audit and Accountability: Policies provide a basis for auditing network activities, ensuring accountability, and supporting investigations when needed.

# Network Scanning Report Generation

Generating a network scanning report is a critical step in the vulnerability management process. The report provides a comprehensive overview of the security posture of a network, detailing identified vulnerabilities, their severity, and recommendations for remediation.

# Network Penetration Testing

1. Reconnaissance → Port Scanning, OS Fingerprinting, Service Fingerprinting
2. Vulnerability Assessment
3. Manual Penetration Testing
4. Report Generation

# Steps

Scan Results Overview:
Provide an executive summary that includes the purpose of the scan, the scope of the assessment, and high-level findings.
Summarize the overall risk posture of the network.

Scope and Objectives:
Clearly define the scope of the scanning activity, including IP ranges, systems, and applications that were assessed.
Outline the objectives of the scan, such as compliance checks or specific vulnerability assessments.

Methodology:
Describe the scanning methodology used, including the tools and techniques employed.
Specify whether authenticated or unauthenticated scans were performed.

Scan Configuration:
Provide details on the configuration settings used during the scan, such as scan profiles, timing, and any specific parameters.

Executive Summary:
Summarize the key findings in a non-technical language for executive stakeholders.
Highlight critical vulnerabilities and potential risks to the organization.

Vulnerability Analysis:
Categorize vulnerabilities based on their severity (e.g., critical, high, medium, low).
Include detailed information about each vulnerability, including CVE identifiers, descriptions, and risk assessments.

Recommendations:
Provide clear and actionable recommendations for remediation.
Prioritize recommendations based on the severity of vulnerabilities and potential impact.

Graphs and Visualizations:
Include graphs and visual representations of the vulnerability distribution, severity levels, and trends over time.
Use charts to illustrate improvements or deteriorations in security posture.

Compliance Status:
If applicable, include a section on compliance checks and the organization's adherence to relevant standards (e.g., PCI DSS, HIPAA).
Highlight any non-compliance issues and provide recommendations for alignment.

Risk Assessment:
Assess the overall risk to the organization based on identified vulnerabilities.
Consider factors such as the likelihood of exploitation and potential impact on business operations.

Mitigation and Remediation Plan:
Outline a comprehensive mitigation and remediation plan.
Provide a timeline for addressing critical and high-risk vulnerabilities.

# Router attacks

Routers are critical components of network infrastructure, responsible for directing data traffic between devices within a network and connecting the network to the internet. Due to their central role, routers can be targeted in various types of attacks.

DNS Spoofing and Cache Poisoning:
DNS Spoofing:
Attackers manipulate the router's DNS settings to redirect users to malicious websites.

DNS Cache Poisoning:
Injecting false DNS information into the router's cache to redirect users or perform man-in-the-middle attacks.

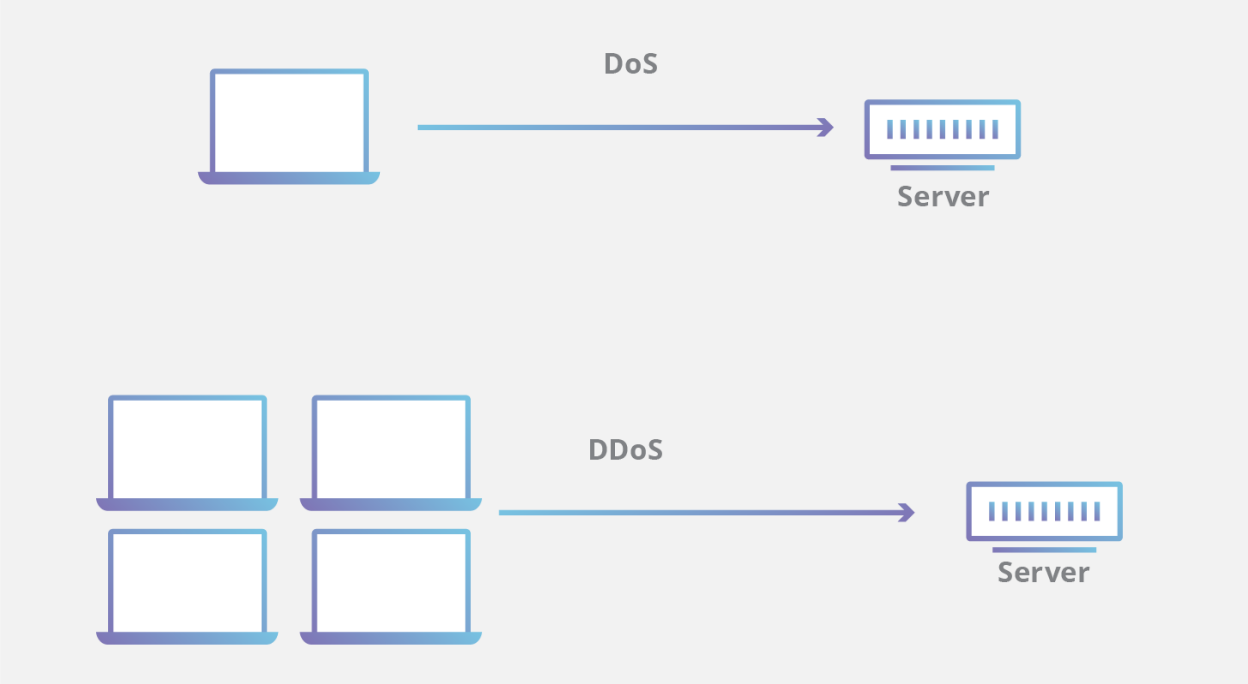
Preventive Measures:
Regularly update router firmware to patch known vulnerabilities.
Use DNS Security Extensions (DNSSEC) to enhance DNS security.
Disable unnecessary DNS services.

Firmware Exploits:
Exploiting vulnerabilities in the router's firmware to gain unauthorized access or control.

Preventive Measures:
Keep router firmware up to date by regularly checking for and applying updates.
Disable remote firmware update capabilities if not required.
Monitor vendor announcements for security patches.

Denial of Service (DoS) Attacks:
Overwhelming the router with traffic to disrupt its normal functioning and render it temporarily or permanently unavailable.

Preventive Measures:
Implement rate limiting and traffic filtering.
Use intrusion prevention systems to detect and mitigate DoS attacks.
Enable features like SYN flood protection.


DoS
Server
DDoS
Server

Man-in-the-Middle (MitM) Attacks:
Intercepting and manipulating data traffic passing through the router to eavesdrop, modify, or inject malicious content.

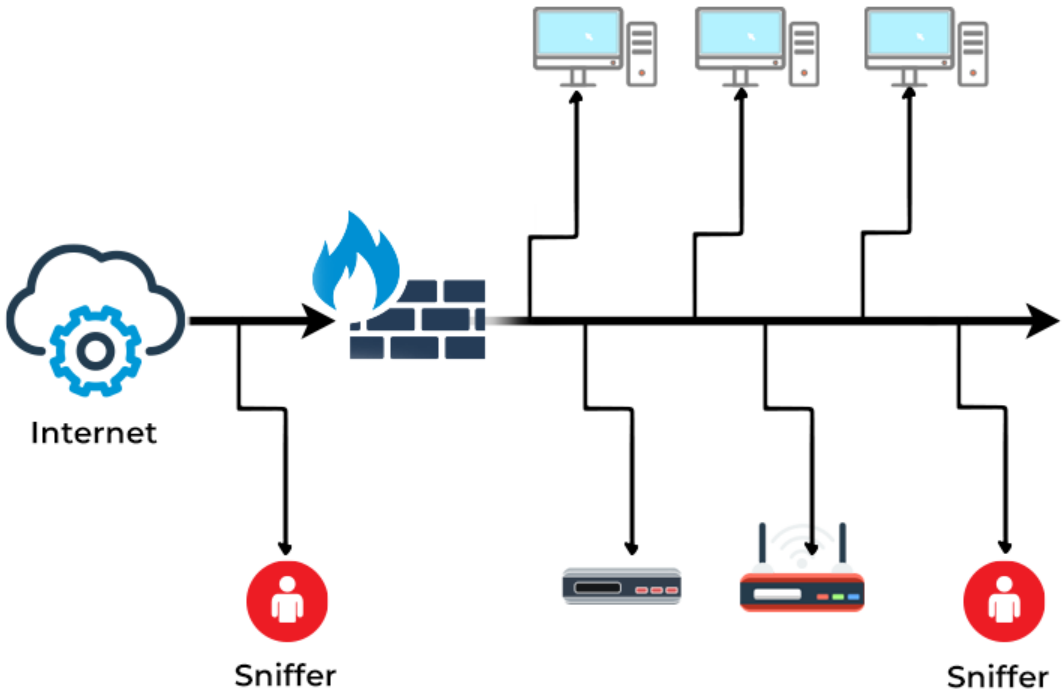
Preventive Measures:
Enable encryption protocols such as WPA3 for Wi-Fi.
Use Virtual Private Network (VPN) technologies for secure communication.
Regularly monitor and audit network traffic for anomalies.

# Packet Sniffing

Packet sniffing, also known as network sniffing or protocol analysis, is the practice of capturing and inspecting data packets as they travel over a computer network. This activity is often done for legitimate purposes, such as network troubleshooting or monitoring, but it can also be misused for malicious activities if performed without proper authorization.

TOOLBOX™

# HOW PACKET SNIFFING ATTACK WORKS



# Components

Data Packets:
In computer networks, data is divided into smaller units called packets. These packets contain information such as source and destination addresses, data payload, and control information.

Network Interface:
Packet sniffing is typically performed at the network interface level, where the network adapter captures packets from the network medium (e.g., Ethernet, Wi-Fi).

# Components

Promiscuous Mode:
Network interfaces operate in either promiscuous or non-promiscuous mode. In promiscuous mode, the interface captures all packets on the network, not just those addressed to the specific device.

# Types of authentication

Password-based Authentication:
Description: Users provide a unique password that is compared to a stored, predefined value.
Strengths: Simple, widely used, and cost-effective.
Weaknesses: Vulnerable to password cracking, phishing, and password reuse.

# Types of authentication

Multi-Factor Authentication (MFA):
Description: Requires users to provide two or more authentication factors (e.g., password, biometric data, security token) to access a system.
Strengths: Enhances security by adding additional layers of authentication.
Weaknesses: Implementation complexity, potential inconvenience for users.

Biometric Authentication:
Description: Uses unique physical or behavioral characteristics of an individual (e.g., fingerprint, iris, voice) for identity verification.
Strengths: Difficult to forge, convenient for users.
Weaknesses: Biometric data can be stolen (although less easily than passwords), and false positives/negatives can occur.

Token-based Authentication:
Description: Involves the use of physical or virtual tokens (e.g., security tokens, smart cards) to generate one-time passwords.
Strengths: Provides dynamic and time-sensitive codes, reducing vulnerability to replay attacks.
Weaknesses: Cost of token deployment, potential loss or theft of tokens.

Certificate-based Authentication:
Description: Relies on digital certificates issued by a trusted Certificate Authority (CA) to verify the identity of users or systems.
Strengths: Strong security, scalable, and suitable for network communication.
Weaknesses: Management overhead in handling and renewing certificates.

Single Sign-On (SSO):
Description: Allows users to authenticate once and access multiple services or applications without re-entering credentials.
Strengths: Improves user experience, reduces password fatigue.
Weaknesses: Security risks if the single set of credentials is compromised.

Risk-based Authentication:
Description: Analyzes user behavior, location, and other contextual factors to determine the risk level and adjust authentication requirements accordingly.
Strengths: Adapts to varying risk levels, providing a balance between security and user experience.
Weaknesses: Requires effective risk analysis and may generate false positives/negatives.

Time-based One-Time Passwords (TOTP):
Description: Generates one-time passwords that are valid for a short period, often using a time-based algorithm.
Strengths: Dynamic and time-sensitive, reducing vulnerability to replay attacks.
Weaknesses: Requires synchronization between the authentication server and the client device.

Knowledge-based Authentication (KBA):
Description: Users answer questions or provide information only they should know, such as personal details or secret questions.
Strengths: Simple and user-friendly.
Weaknesses: Vulnerable to social engineering and easily guessable answers.

Adaptive Authentication:
Description: Adjusts the level of authentication based on user behavior, risk factors, and contextual information.
Strengths: Provides a dynamic and context-aware approach to authentication.
Weaknesses: Requires effective risk analysis and may require more complex implementations.

## Fake Authentication Attack

A fake authentication attack, also known as a "spoofed authentication" or "forged authentication" attack, involves an adversary attempting to gain unauthorized access to a network or system by pretending to be an authenticated user or device. This type of attack often exploits vulnerabilities in the authentication process, relying on deceptive techniques to trick the target system into accepting the fake credentials.

# De authentication

De authentication, often referred to as "deauth," is a network management frame in Wi-Fi networks that is used to disconnect a device from a Wi-Fi network. While de authentication itself is a legitimate and useful feature for network administrators to manage and control access to a Wi-Fi network, it can also be misused for malicious purposes.

# Attacks on WPA and WPA-2 Encryption

WPA (Wi-Fi Protected Access) and WPA2 are wireless encryption protocols designed to secure Wi-Fi networks. However, like any security protocol, they are not immune to attacks.

# Attacks on WPA and WPA2 encryption:

Brute Force Attacks:
Description: Attackers attempt to guess the pre-shared key (PSK) or passphrase by systematically trying all possible combinations.
Prevention: Use strong, complex passwords to increase resistance to brute force attacks. Implementing WPA3, where available, is recommended as it includes improved security features.

# Attacks on WPA and WPA2 encryption:

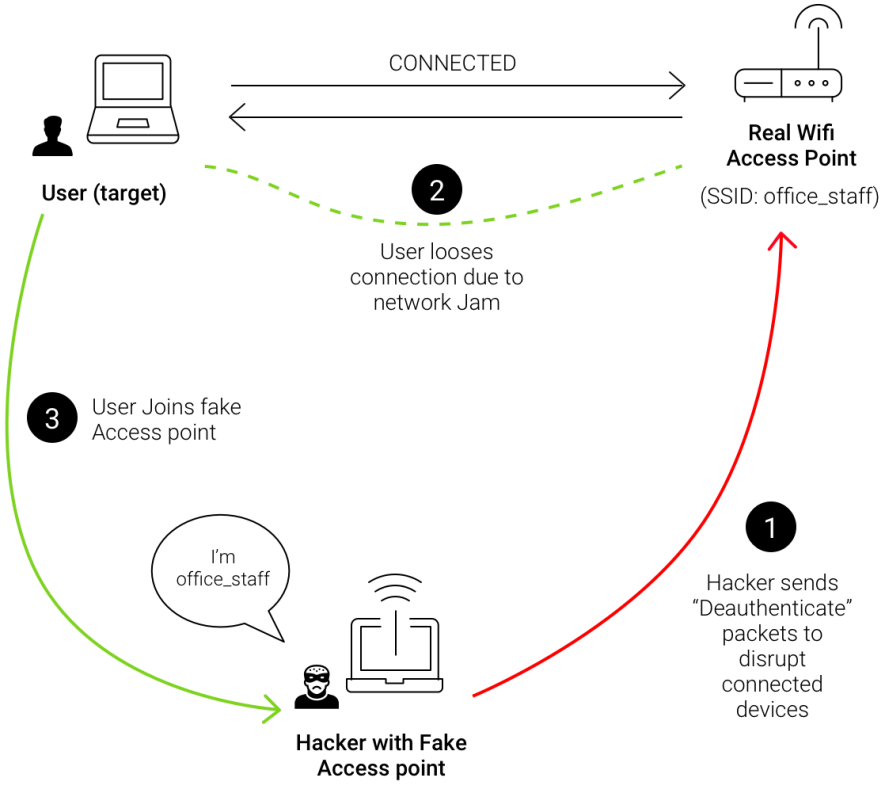
Dictionary Attacks:
Description: Attackers use precompiled lists of common passwords (dictionaries) to attempt to crack the passphrase.
Prevention: Choose unique and complex passphrases that are less likely to be found in common dictionaries. Use a passphrase with a mix of uppercase, lowercase, numbers, and symbols.

WPS (Wi-Fi Protected Setup) Attacks:
Description: Exploiting vulnerabilities in WPS, attackers can attempt to guess the PIN used for authentication.
Prevention: Disable WPS on the router if not needed. Use a long and random WPS PIN if WPS is necessary.

Evil Twin Attacks:
Description: Attackers set up rogue Wi-Fi networks with names similar to legitimate networks, tricking users into connecting to the malicious network.
Prevention: Educate users about the risks of connecting to unknown networks. Use features like WPA3-SAE (Simultaneous Authentication of Equals) for improved security.

# Fake hotspot

A fake hotspot, also known as a rogue hotspot or evil twin, is a type of cyber attack where an attacker sets up a malicious Wi-Fi access point to trick users into connecting to it, thinking it is a legitimate and trusted network. This type of attack is a form of deception and is commonly used for various malicious purposes, including eavesdropping on user activities, capturing sensitive information, and launching man-in-the-middle attacks.


CONNECTED
Real Wifi
Access Point
(SSID: office_staff)
User (target)
2
User looses
connection due to
network Jam
3
User Joins fake
Access point
I'm
office_staff
1
Hacker sends
"Deauthenticate"
packets to
disrupt
connected
devices
Hacker with Fake
Access point

# WPA & WPA-2 attacks

WPA (Wi-Fi Protected Access) and WPA2 are widely used wireless security protocols designed to secure Wi-Fi networks. However, like any security mechanism, they are susceptible to various attacks.

# Common attacks targeting WPA and WPA2:

Brute Force Attacks:
Description: Attackers attempt to guess the pre-shared key (PSK) or passphrase by systematically trying all possible combinations.
Prevention: Use strong, complex passwords and passphrases. Implementing WPA3, where available, is recommended for enhanced security.

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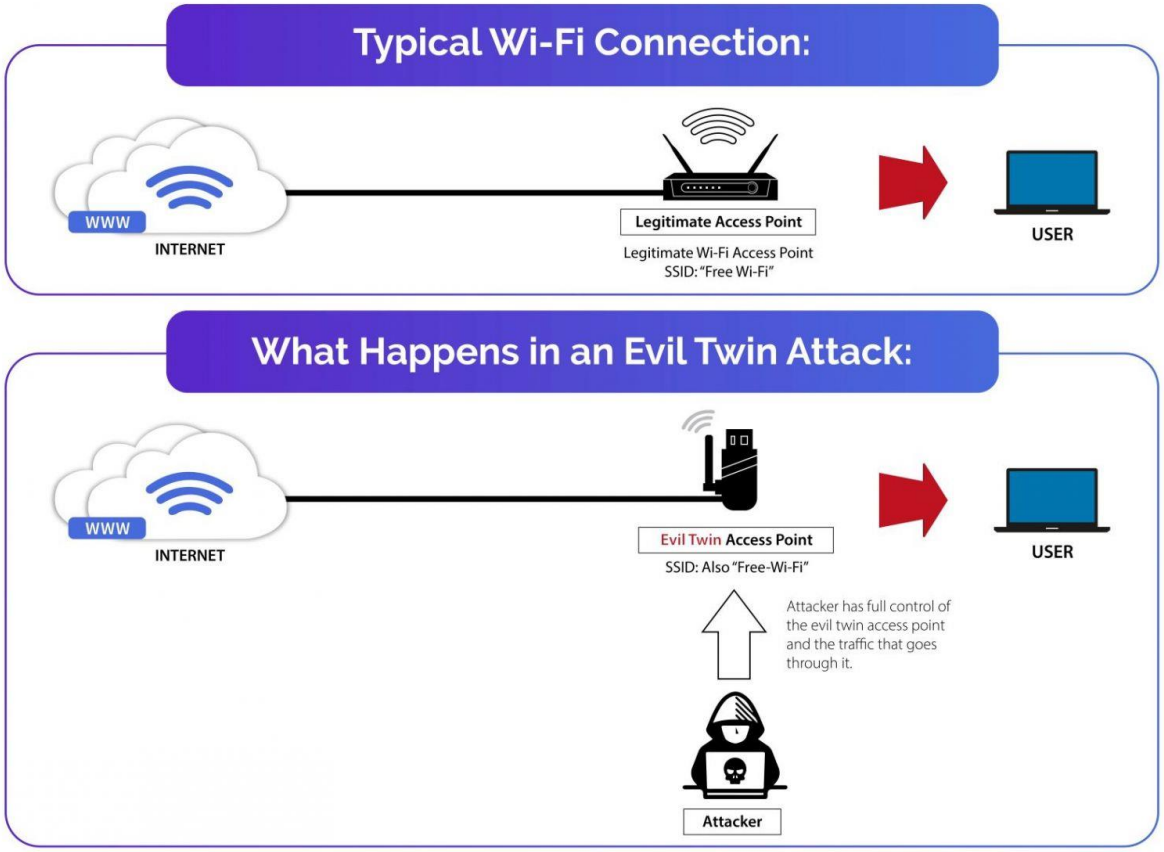
Typical Wi-Fi Connection:
WWW
INTERNET
Legitimate Access Point
Legitimate Wi-Fi Access Point
SSID: "Free Wi-Fi"
USER
What Happens in an Evil Twin Attack:
WWW
INTERNET
Evil Twin Access Point
SSID: Also "Free-Wi-Fi"
USER
Attacker has full control of the evil twin access point and the traffic that goes through it.
Attacker

# Wireless Hacking using phishing

Wireless hacking using phishing involves exploiting human vulnerabilities to gain unauthorized access to a wireless network. Phishing is a social engineering technique where attackers deceive individuals into revealing sensitive information, such as login credentials or personal data. In the context of wireless networks, phishing attacks may target Wi-Fi passwords or attempt to compromise security.

# Common methods used in wireless hacking through phishing:

Fake Wi-Fi Hotspots:
Description: Attackers set up rogue Wi-Fi hotspots with names similar to legitimate networks. Users unknowingly connect to these fake hotspots, allowing attackers to intercept and manipulate their traffic.
Prevention: Verify the legitimacy of Wi-Fi networks before connecting. Avoid connecting to open or unsecured networks, especially those with names similar to well-known public hotspots.

# Common methods used in wireless hacking through phishing:

Evil Twin Attacks:
Description: Attackers create a malicious Wi-Fi access point with a name (SSID) similar to a legitimate network. Users may connect, thinking it's a trusted network, enabling attackers to intercept traffic.
Prevention: Be cautious when connecting to networks with names similar to known networks. Confirm the legitimacy of the network with administrators.

Phishing Portals and Captive Portals:
Description: Attackers create fake login pages resembling those of legitimate networks. When users connect to a rogue Wi-Fi network, they may be redirected to a phishing portal, where their login credentials are captured.
Prevention: Avoid entering sensitive information on unsecured websites. Ensure the legitimacy of login pages by checking for secure connections (https://) and valid certificates.

Credential Harvesting:
Description: Attackers send phishing emails or messages pretending to be from legitimate sources, tricking users into providing Wi-Fi login credentials.
Prevention: Be skeptical of unsolicited emails or messages. Verify the legitimacy of communication with the supposed sender before sharing any sensitive information.

# MITM (man in the middle attack)

A Man-in-the-Middle (MitM) attack is a type of cyber attack where an unauthorized third party intercepts and potentially alters the communication between two parties without their knowledge. The attacker positions themselves between the communicating parties, gaining the ability to eavesdrop on or manipulate the data being exchanged. MitM attacks can occur in various contexts, including network communication, Wi-Fi connections, and secure communication channels.

# Brute Force Attacks

A brute force attack is a trial-and-error method used by attackers to gain access to a system, network, or application by systematically trying all possible combinations of usernames and passwords. The goal of a brute force attack is to discover the correct login credentials that allow unauthorized access. This type of attack is straightforward but can be time-consuming, especially for complex and lengthy passwords.

# Brute Force Attacks Explained

In a brute force attack, a cybercriminal uses trial and error to try and break into a device, network, or website.

An attacker **utilizes a hacking tool.**

The hacking tool **attempts multiple logins.**

The system **returns a valid or invalid response.**

# Characteristics of Brute Force Attacks:

Repetitive Attempts:
Brute force attacks involve repetitive attempts to guess login credentials by trying all possible combinations. Attackers automate the process using software or scripts.

Resource Intensive:
Brute force attacks can be resource-intensive, requiring significant computational power and time, especially when dealing with strong passwords.

# Characteristics of Brute Force Attacks:

Dictionary Attacks:
Brute force attacks often include dictionary attacks, where attackers use a predefined list of common passwords or words to speed up the process.

Credential Stuffing:
In credential stuffing attacks, attackers reuse username-password pairs obtained from previous data breaches, testing them against various online accounts.

Lockout Policies:
Attackers may attempt to bypass lockout policies by using multiple IP addresses or compromising a large number of user accounts simultaneously.